Global Financial Cycle

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Cycle in asset prices and capital flows

- asset prices correlated across countries
- financial flows correlated across countries
- asset prices and financial flows correlated in time series
- lots of cross-sectional heterogeneity

basic facts

One dominant component explains 25% of variation in global asset prices

Two principal components explain 35% of variation in financial flows

- first component strongly related to the dominant component in prices
- first component strongly related to measures of risk-taking capacity
- second component related to trade flows

how to model the GFC

How to model?

One approach: shocks that lead to risk repricing

- why would repricing happen?
- what kind of risk?
- how to get financial flows and asset prices move correctly?
- how to get realistic cross-sectional heterogeneity?

today

Two models: based on idiosyncratic and global risk

Model with idiosyncratic risk:

- focus on diversification frictions
- cross-sectional heterogeneity in price and flow responses
- gross flows are the mechanism that generates the right cross-section

Model with global risk:

- focus on global factors behind currencies
- look at global drivers of exchange rates
- exposure to real shocks generates exposure to financial shocks

GFC: idiosyncratic risk and gross flows

why idiosyncratic risk?

Dominant components capture much of variation, but

- the share of global factors is lower for emerging markets
- asset prices and financial flows are more volatile in emerging markets
- emerging markets are especially prone to sudden stops, currency depreciation

What if we drop the assumption that idiosyncratic risk if diversifiable?

why gross flows?

What is special about gross flows:

- larger and more volatile than net flows
- retrenchment of local investors can compensate for foreign investors leaving

An insight originally from Caballero and Simsek (2020)

- show benefits of having external assets
- operationalize retrenchment as a stabilization mechanism
- study optimal financial openness

outline of the model

Minimal set of ingredients to determine gross flows and returns in cross-section:

- a set of countries: asset + local investor
- assets subject to country-specific shocks
- financial market segmentation: global intermediaries with limited capacity

Two sets of experiments:

- idiosyncratic shocks under imperfect diversification: local capital flight
- aggregate shocks to pricing of local risk: global capital flight

countries

Countries $i \in [0, 1]$

- Lucas tree with price p_{it}, fixed supply of 1
- cumulative dividend up to t denoted by y_{it}
- flow dividend $dy_{it} = \nu dt + \sigma dZ_{it}$

problem of local agents

$$\max_{\{c_{it},\theta_{it}\}_{t\geq 0}} \mathbb{E}\left[\rho \int_0^\infty e^{-\rho t} \ln(c_{it}) dt\right] \tag{1}$$

$$dw_{it} = (r_t w_{it} - c_{it})dt + \theta_{it} w_{it} dR_{it} \tag{2}$$

- allocate share θ_{it} to tree
- share $1 - \theta_{it}$ to intermediary's debt, interest rate $r_t dt$

Excess returns dR_{it} are given by

$$dR_{it} = \frac{1}{p_{it}}(dy_{it} + dp_{it}) - r_t dt \equiv \mu_{it}^R dt + \sigma_{it}^R dZ_{it} \tag{3}$$

special country

Special country

- Lucas tree with price $\hat{p}_t$, fixed supply of $\hat{q}$
- cumulative dividend up to t denoted by $\hat{y}_t$
- flow dividend $d\hat{y}_t = \nu dt$

Excess returns $d\hat{R}_t$ given by

$$d\hat{R}_t = \frac{1}{\hat{p}_t}(\nu dt + d\hat{p}_t) - r_t dt \tag{4}$$

Houses the global intermediary

intermediary

Objective and budget constraint:

$$\max_{\{\hat{c}_t, \hat{\boldsymbol{\theta}}_t, \hat{\theta}_t\}_{t\geq 0}} \mathbb{E}\left[\rho \int_0^\infty e^{-\rho t} \ln(\hat{c}_t) dt\right] \tag{5}$$

$$d\hat{w}_t = (r_t \hat{w}_t - \hat{c}_t)dt + \int (\hat{\theta}_{it} \hat{w}_t dR_{it}) di + \hat{\theta}_t \hat{w}_t d\hat{R}_t \tag{6}$$

- portfolio shares $\{\hat{\theta}_{it}\}$ allocated to all trees

Constraint on total amount of idiosyncratic risk:

$$\underbrace{\int_0^1 \mathbb{V}_t[\hat{\theta}_{it} \hat{w}_t dR_{it}] di}_{\text{total idiosyncratic risk}} \leq \gamma_t \hat{w}_t \underbrace{\int_0^1 \mathbb{E}_t[\hat{\theta}_{it} \hat{w}_t dR_{it}] di}_{\text{expected profit}} \tag{7}$$

holdings

- tree holdings: $h_{it} = \dfrac{\theta_{it} w_{it}}{p_{it}}$ and $\hat{h}_{it} = \dfrac{\hat{\theta}_{it} \hat{w}_t}{p_{it}}$
- special tree holdings: $\hat{s}_t = \dfrac{\hat{\theta}_t \hat{w}_t}{\hat{p}_t}$
- bond holdings: $b_{it} = (1 - \theta_{it}) w_{it}$ and $\hat{b}_t = \left(1 - \int \hat{\eta}_{it} - \hat{\theta}_t\right) \hat{w}_t$

equilibrium

Definition: processes for prices $\{p_{it}, \hat{p}_t, r_t\}$, quantities $\{c_{it}, \hat{c}_t, \hat{h}_{it}, \hat{h}_{it}, \hat{s}_{it}, b_{it}, \hat{b}_{it}\}$, wealth $\{w_{it}, \hat{w}_t\}$ such that all agents optimize and the following markets clear:

world map

$$1 = \hat{h}_{it} + h_{it} \qquad \text{all } i \in [0,1] \tag{8}$$

$$\hat{q} = \int_0^1 \hat{s}_{it} di \tag{9}$$

$$0 = \int_0^1 \hat{b}_{it} di + \int_0^1 b_{it} di \tag{10}$$

$$(1+\hat{q})\nu = \int_0^1 \hat{c}_{it} di + \int_0^1 c_{it} di \tag{11}$$

main equation

$$\mu_{it}^{p} + \nu - r_t p_{it} = \underbrace{(\sigma_{it}^{p} + \sigma)^2}_{\text{quantity of risk}} \cdot \underbrace{\frac{1}{w_{it} + \gamma_t \hat{w}_t}}_{\text{price of risk}} \tag{12}$$

analytical solutions: prices

Consider a second-order approximation around $\sigma = 0$ (older terms are $o(\sigma^3)$, in fact)

- define price of risk function $\boldsymbol{\pi}(w) = \dfrac{1}{w + \gamma \hat{w}}$
- strength of precautionary motives relates to average price of risk:

$$r = \rho - \frac{\rho\sigma^2}{\nu} \cdot \frac{\mathbb{E}[\boldsymbol{\pi}(w)]}{1 + \hat{q}} \tag{13}$$

- asset prices reflect risk premia and precautionary motives:

$$p(w) = \frac{\nu}{\rho} + \frac{\sigma^2}{\rho} \cdot \left(\frac{\mathbb{E}[\boldsymbol{\pi}(w)]}{1 + \hat{q}} - \boldsymbol{\pi}(w) \right) \tag{14}$$

$$\hat{p} = \frac{\nu}{\rho} + \frac{\sigma^2}{\rho} \cdot \frac{\mathbb{E}[\boldsymbol{\pi}(w)]}{1 + \hat{q}} \tag{15}$$

analytical solutions: wealth distribution

Limiting steady-state distribution non-degenerate

- drift proportional to σ^2, volatility proportional to σ:

$$\frac{dw}{w} = \left(\boldsymbol{\pi}(w)^2 - \frac{\rho}{\nu} \cdot \frac{\hat{\mathbb{E}}[\boldsymbol{\pi}(w)]}{1+\hat{q}} \right) \sigma^2 dt + \boldsymbol{\pi}(w)\sigma dZ + o_p(\sigma^2) \tag{16}$$

- volatility approaches zero at infinity
- drift is negative for large w, positive for $w \longrightarrow 0$ if γ not too large
- risky countries grow, safe countries consume away

Shock to risk-taking capacity

shock to risk-taking capacity γ

Suppose $\gamma(t)$ falls

At steady-state prices

- intermediaries would decrease portfolio shares equally
- hold more in poor countries $\longrightarrow$ will want to sell more

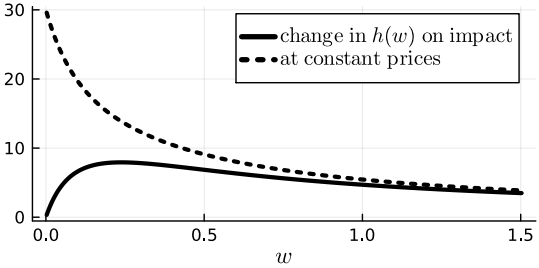
Local agents retrench:

- large volumes stabilize risk premia in rich countries
- agents in poor countries cannot absorb much without a sharp rise in risk premia

change in holdings on impact

Change in domestic holdings $h(w)$ on impact (in percent of total supply)

- counterfactual, at constant steady-state prices
- actual, in equilibrium



change in prices on impact

Price changes on impact: responses to interest rate $r(t)$ and to global factor $\varphi(t) = \gamma(t)\hat{w}(t)$:

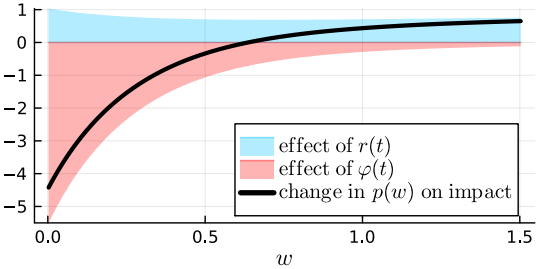


Figure: percentage changes in $p(w, t)$ on impact.

Empirical evidence

cross-sectional regressions

Loadings $\{\beta_i^{(r)}\}_{i\in\mathcal{I}}$ from the factor regression with ouctome r_{it} and f_t

$$r_{it} = \alpha^{(r)} + \beta_i^{(r)} f_t + \epsilon_{it}^{(r)} \tag{17}$$

Project the loadings $\{\beta_i^{(r)}\}$ on cross-sectional variable x_i (external assets over GDP):

$$\beta_i^{(r)} = \gamma^{(r)} + \Gamma^{(r)} x_i + \varepsilon_i^{(r)} \tag{18}$$

- <u>outcomes</u> r_{it}: USD returns on stocks, outward flows (normalized by assets and by liabilities)
- <u>factor</u> f_t: first principal component in outward flows

return on equities

Table: cross-sectional regressions of loadings from r_{it}^{msci} on assets-to-GDP

	$\beta^{(msci)}$
Γ	**-0.0127**
	(0.0050)
N	42
R^2	0.081

asset acquisition (normalized by assets)

Table: cross-sectional regressions of loadings from $r_{it}^{acq/asset}$ on assets-to-GDP

	$\beta^{(acq/asset)}$
Γ	**0.0103**
	(0.0052)
N	58
R^2	0.056

asset acquisition (normalized by liabilities)

Table: cross-sectional regressions of loadings from $r_{it}^{acq/liab}$ on assets-to-GDP

	$\beta^{(acq/liab)}$
Γ	**0.0129**
	(0.0027)
N	57
R^2	0.243

correlation between loadings

Run the following projections:

$$\beta_i^{(msci)} = \kappa^{(msci,acq/asset)} + K^{(msci,acq/asset)} \beta_i^{(acq/asset)} + \varepsilon_i^{(msci,acq/asset)} \tag{19}$$

$$\beta_i^{(msci)} = \kappa^{(msci,acq/liab)} + K^{(msci,acq/liab)} \beta_i^{(acq/liab)} + \varepsilon_i^{(msci,acq/liab)} \tag{20}$$

This shows how asset return cyclicality relates to financial flow cyclicality

loadings in cross-section

Table: cross-sectional regressions of loadings of r_{it}^{msci} on loadings of $r_{it}^{acq/asset}$ and $r_{it}^{acq/liab}$

	$\beta^{(acq/asset)}$	$\beta^{(acq/liab)}$
K	-0.160	**-0.277**
	(0.192)	(0.077)
N	29	29
R^2	0.034	0.179

takeaways from empirical analysis

Countries with higher assets-to-GDP ratio have less cyclical equity returns

- advanced economies (higher asset ratios) outperform emerging markets in bad times

Countries with higher assets-to-GDP ratio have more cyclical outward flows

- advanced economies (higher asset ratios) retrench more in bad times
- this is especially pronounced in retrenchment relative to liabilities

Retrenchment relative to liabilities explains return cyclicality

GFC: real shocks, financial shocks, and the cross-section of currencies

why global shocks?

Factor structure of currencies:

- two factors capture close to half of variation on average
- points to undiversifiable aggregate risk
- should be taken seriously in a model with risk-taking shocks

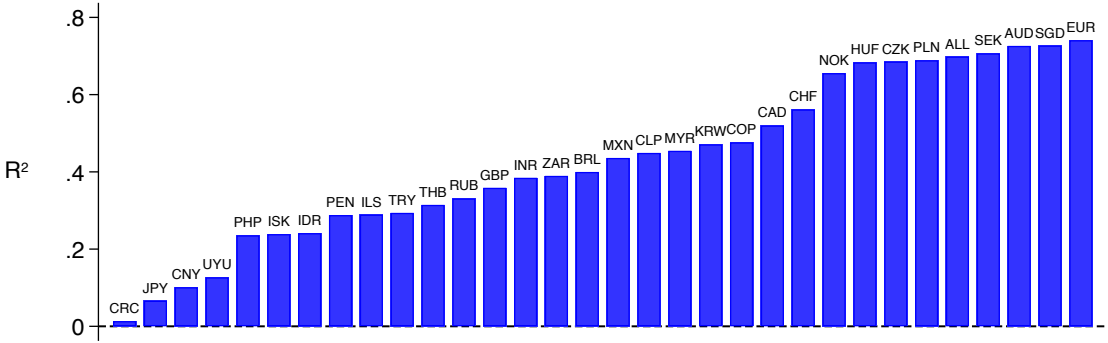
Real and financial shocks both important in cross-section

- heterogeneous exposure to real shocks leads to heterogeneous exposure to financial shocks

Factor structure

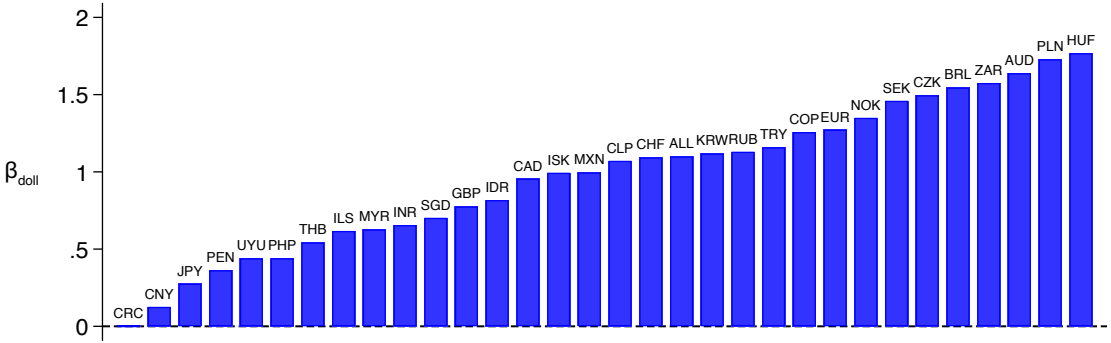
explanatory power

$$\Delta s_{i,t+1} = \beta_{i,\text{comm}}\text{comm}_{t+1} + \beta_{i,\text{doll}}\text{doll}_{t+1} + \varepsilon_{i,t+1}$$



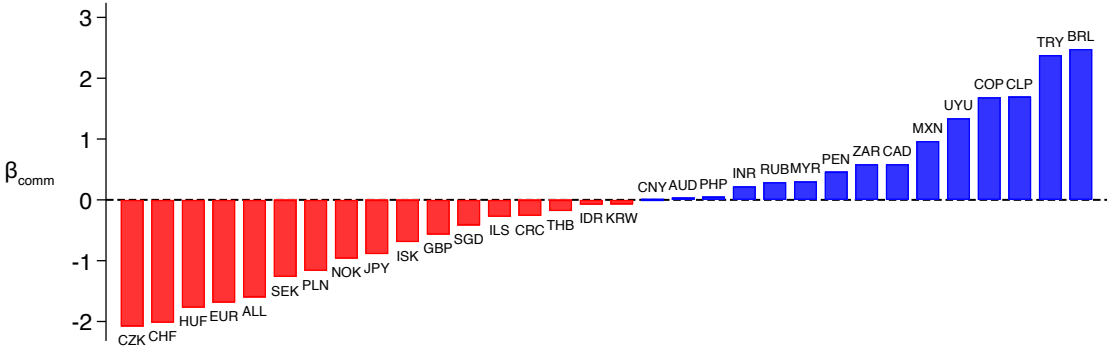
dollar loadings

$$\Delta s_{i,t+1} = \beta_{i,\text{comm}}\text{comm}_{t+1} + \beta_{i,\text{doll}}\text{doll}_{t+1} + \varepsilon_{i,t+1}$$

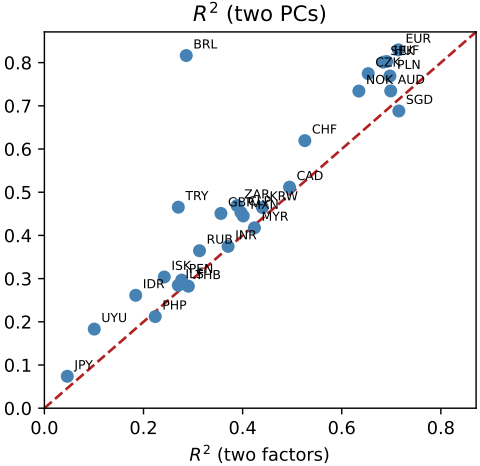


commodity loadings

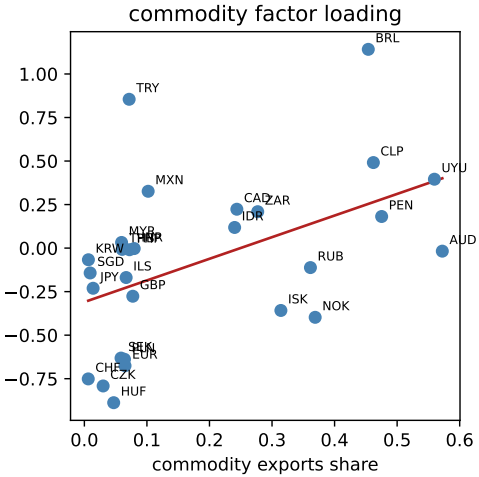
$$\Delta s_{i,t+1} = \beta_{i,\text{comm}} \text{comm}_{t+1} + \beta_{i,\text{doll}} \text{doll}_{t+1} + \varepsilon_{i,t+1}$$

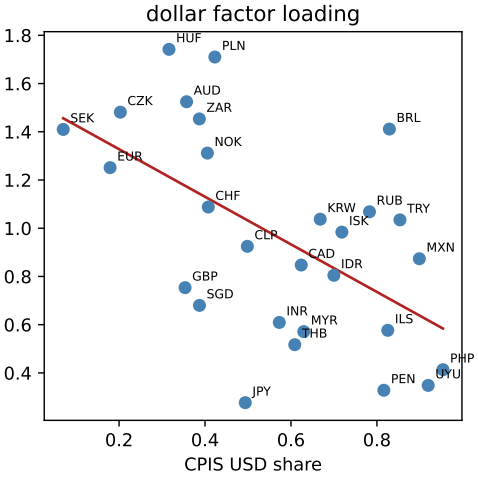


benchmarking explanatory power



determinants of loadings





Model

outline

A continuum of countries + the US

- one tradable good, supply shock z_t, individual exposures e_i
- one non-traded good per country, supply shock x_{it}
- reserve management rule: dollar bonds, holdings centered around M_i, respond to USD

Intermediary:

- long-short portfolio, invests in all currencies
- SDF combines all countries + additional curvature γ_t
- SDF exposed to aggregate shocks z_t and x_{ut}

factor structure in the model

Define depreciation vs the USD as $\Delta s_{i,t+1}$. Excess returns load on aggregate and local shocks:

$$\Delta s_{i,t+1} = \underbrace{\theta(e_i - \bar{e})\Delta z_{t+1} + \xi(M_i - \bar{M})\Delta x_{u,t+1}}_{\text{global real shocks}} + \underbrace{\mu_i \Delta\gamma_{t+1}}_{\text{GFC}} + \alpha\Delta x_{i,t+1} \tag{21}$$

Intermediary's SDF is

$$\lambda_{t+1} = \kappa_t - (\Lambda_z \Delta z_{t+1} + \Lambda_x \Delta x_{u,t+1}) \tag{22}$$

The GFC component in each currency is

$$\mu_i \propto \mathbb{C}_t[\Delta s_{i,t+1}, \lambda_{t+1}] = -\theta(e_i - \bar{e})\Lambda_z \sigma_z^2 - \xi(M_i - \bar{M})\Lambda_x \sigma_x^2 \tag{23}$$

Plugging back, get a two-factor structure:

$$\Delta s_{i,t+1} = \theta(e_i - \bar{e})F_{1,t+1} + \xi(M_i - \bar{M})F_{2,t+1} + \alpha\Delta x_{i,t+1} \tag{24}$$

what are these factors?

The commodity factor slices the cross-section along exposure to commodities:

$$F_{1,t+1} = 2\int_{e_i > \bar{e}} \Delta s_{i,t+1} di - 2\int_{e_i < \bar{e}} \Delta s_{i,t+1} di \tag{25}$$

This uses independence of (e_i, M_i)

The dollar factor is the average:

$$F_{2,t+1} = \int \Delta s_{i,t+1} di \tag{26}$$

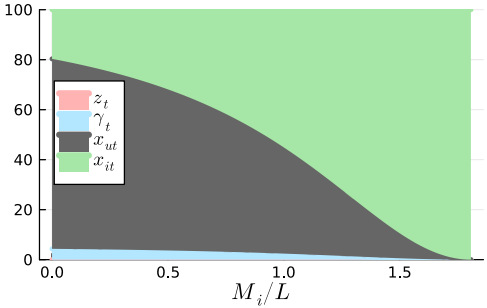
This uses $\int \theta(e_i - \bar{e}) di = 0$ and $\int \xi(M_i - \bar{M}) di = -1$

what drives the factors?

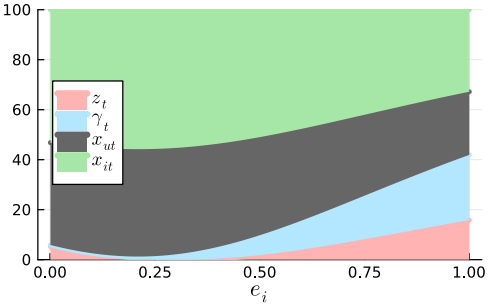
Table: variance shares

	doll_t	comm_t
$\Delta\gamma_t$	41%	26%
Δz_t		74%
Δx_{ut}	59%	

drivers of exchange rates in cross-section



(a) shares of exchange rate variance, $e_i = e$



(b) shares of exchange rate variance, $M_i = L$

Thanks!

references I

Caballero, R. J. and A. Simsek (2020). A model of fickle capital flows and retrenchment. *Journal of Political Economy 128*(6), 2288–2328.